

Lab Assignment #05

Aim:

Design a horn antenna at a given resonant frequency and analyze the results.

Theory:

A horn antenna is a flared waveguide that functions as an impedance transformer, providing a gradual transition between the enclosed feed and free space. This structural expansion smoothly matches the propagating mode's characteristic impedance to the intrinsic impedance of free space, minimizing wave reflection and maximizing radiation efficiency.

Pyramidal horns flare outward in both the E-plane and H-plane, confining the electromagnetic wave to produce a highly directive, narrow beam. While maximum directivity is proportional to the physical aperture area, the natural spherical phase taper of the wavefront across the opening reduces the overall aperture efficiency.

The design of the pyramidal horn antenna begins by establishing the operating parameters within the standard WR137 waveguide frequency band of 5.85 to 8.20 GHz. For this simulation, a center frequency of 7 GHz was selected. The fundamental free-space wavelength, λ , is determined by the ratio of the speed of light to this chosen frequency. To achieve a target theoretical gain of 18 dBi, the required physical aperture area, A , is calculated using the standard antenna gain equation. Assuming a theoretical 100 percent aperture efficiency for the initial baseline calculation, rearranging this formula yields a required aperture area of 8857.64 square millimeters.

$$\lambda = \frac{c}{f}$$

$$G = \frac{4\pi A}{\lambda^2} \implies A = \frac{G\lambda^2}{4\pi} = 8857.64 \text{ mm}^2$$

With the total aperture area established, the specific physical dimensions of the horn can be derived. The broad dimension of the aperture, designated as a_x , was predefined as 120 mm. The narrow dimension, a_y , is then calculated geometrically by dividing the total area by this broad dimension. To ensure a gradual impedance transition from the rectangular waveguide to free space and to minimize phase error across the radiating aperture, the half-flare angle is constrained to a maximum of 15 degrees.

$$a_y = \frac{A}{a_x} = 73.81 \text{ mm}$$

The required axial length to accommodate the broad dimension flare, L_a , is calculated using a trigonometric relationship. This involves the aperture width, the standard broad dimension of the feeding WR137 waveguide ($w_x = 34.8488 \text{ mm}$), and the tangent of the maximum half-flare angle. By evaluating the required flare lengths for both the E-plane and H-plane and taking the maximum to strictly maintain the 15-degree constraint, the final overall length of the horn, a_l , is established.

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$$L_a = \frac{a_x - w_x}{2 \tan(15^\circ)} = 158.89 \text{ mm}$$

These derived physical dimensions were then modeled and simulated across the recommended frequency band. The optimal operating point of the antenna was identified by evaluating the S11 reflection coefficient, with the point of least reflection observed at 7.2248 GHz. Furthermore, farfield simulation revealed an observed directivity of 16.94 dBi. This natural reduction from the theoretical 18 dBi target translates to an actual aperture efficiency of approximately 78 percent, which is a standard characteristic of pyramidal horns caused by the spherical phase taper across the physical aperture.

Table 1 - Parametric Details of WR-137

Parameter	Value
Broad Dimension (Width)	34.8488 mm
Narrow Dimension (Height)	15.7988 mm
Cutoff Frequency of Lowest Order Mode	4.301 GHz
Cutoff Frequency of Upper Mode	8.603 GHz
Recommended Frequency Band	5.85 to 8.20 GHz
Broad Dimension Aperture Length	120 mm
Narrow Dimension Aperture Length	73.81 mm
Length of Horn	158.89 mm

Procedure:

1. Open CST Studio Suite and select the New Template option from the main dashboard.
2. Select the Microwave & RF / Optical template folder, choose Antennas, and then select Waveguide (Horn).
3. Select the Time Domain Solver from the available solver options and proceed to the next step.
4. Configure the project units by setting Dimensions to mm and Frequency to GHz.
5. Define the Frequency Range with a Minimum of 5.85 GHz and a Maximum of 8.20 GHz to cover the standard operating band for the WR137 waveguide feed.
6. Open the Modeling tab, select Background, and ensure the Material Type is set to Normal with boundaries configured as Open (add space) in all directions to allow the electromagnetic waves to radiate into free space.
7. Open the Parameter List window and create a new parameter named wx with a value of 34.8488 to represent the broad dimension of the feed.
8. Create a second parameter named wy with a value of 15.7988 to represent the narrow dimension of the feed.
9. Create a third parameter named wl with a value of 80 to represent the length of the waveguide feed.
10. Create a fourth parameter named ax with a value of 120 to represent the broad dimension of the radiating aperture.
11. Create a fifth parameter named ay with a value of 73.81 to represent the narrow dimension of the radiating aperture.
12. Create a sixth parameter named al with a value of 158.89 to represent the axial length of the horn flare.
13. Select the Brick tool from the Modeling tab, press Esc to open the numerical entry dialog box, and configure the waveguide feed interior with Xmin as $-wx/2$, Xmax as $wx/2$, Ymin as $-wy/2$, Ymax as $wy/2$, Zmin as $-wl$, and Zmax as 0, assigning the material as Vacuum.
14. Select the Brick tool again, press Esc, and define a thin template for the aperture profile by setting Xmin as $-ax/2$, Xmax as $ax/2$, Ymin as $-ay/2$, Ymax as $ay/2$, Zmin as al, and Zmax as al, assigning the material as Vacuum.
15. Activate the Pick Face tool (shortcut F), select the front face of the waveguide feed located at $Z=0$, select the rear face of the aperture template at $Z=al$, and use the Loft tool to connect them, generating the inner vacuum volume of the horn flare.
16. Select both the waveguide feed brick and the lofted horn flare, use the Boolean Add tool to combine them into a single solid, and apply the Shell tool with an outside thickness of 2 mm and PEC material to generate the conductive metallic walls of the antenna.
17. Navigate to the Simulation tab, click on Field Monitor, and define farfield monitors for your design frequency of 7 GHz and your observed optimal frequency of 7.2248 GHz.

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18. Activate the Pick Face tool, double-click the rear face of the internal vacuum feed located at $Z=-w_l$, click on Waveguide Port in the Simulation tab, and click OK to define the excitation port.
19. Click on Setup Solver in the Simulation tab, verify your settings, and click Start to initialize the Time Domain solver.
20. Navigate to the 1D Results folder in the navigation tree to retrieve the S-Parameter plots and verify the lowest S11 value at 7.2248 GHz.
21. Navigate to the Farfields folder within the 2D/3D Results tree to visualize the 3D radiation pattern and confirm the observed directivity of 16.94 dBi.

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3D Model:

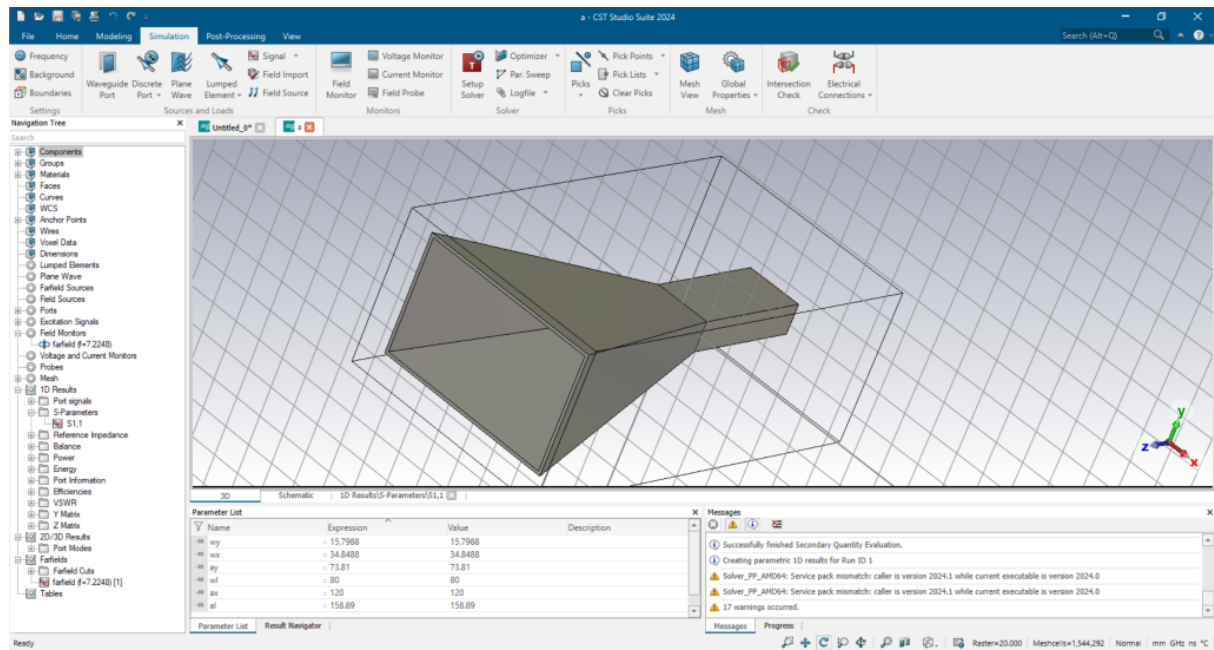


Fig 1.1 3D Model of Horn Antenna

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Results:

Below are the results for simulating EM waves through a WR-137 Horn Antenna over the range of frequencies 5.85 to 8.20 GHz.

Scattering Parameters

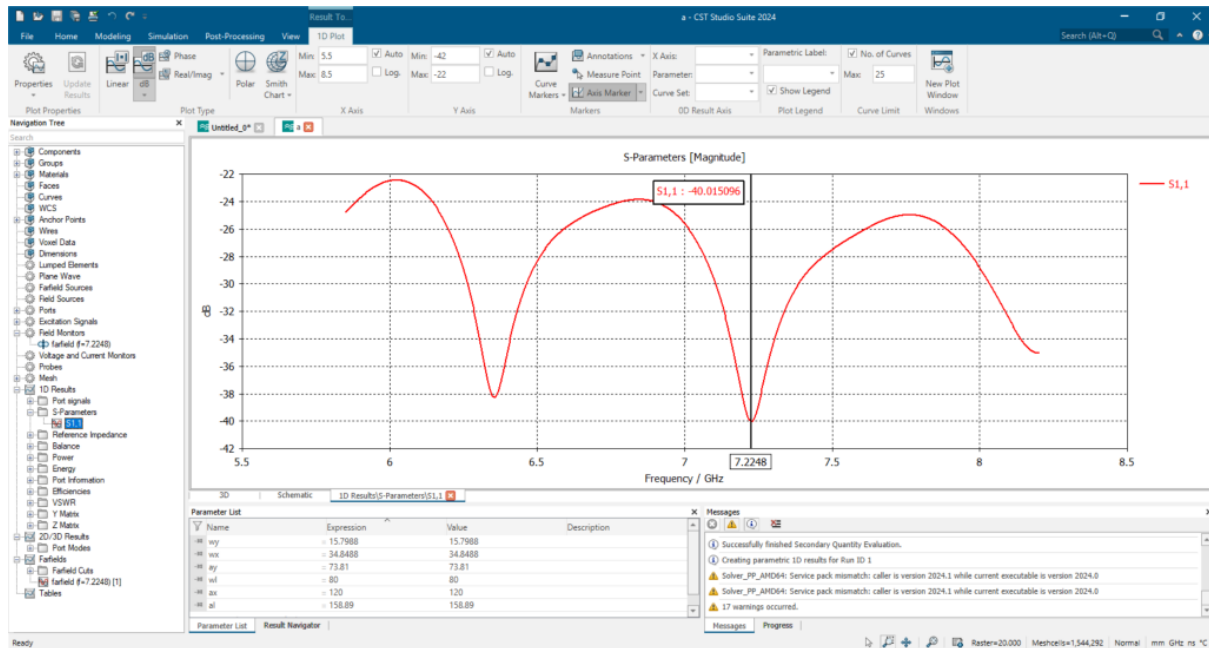


Fig 2.1 Scattering parameter $S(1,1)$, minimum value found at 7.2248 GHz, which will be used to simulate resonant frequency to obtain maximum possible gain

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VSWR

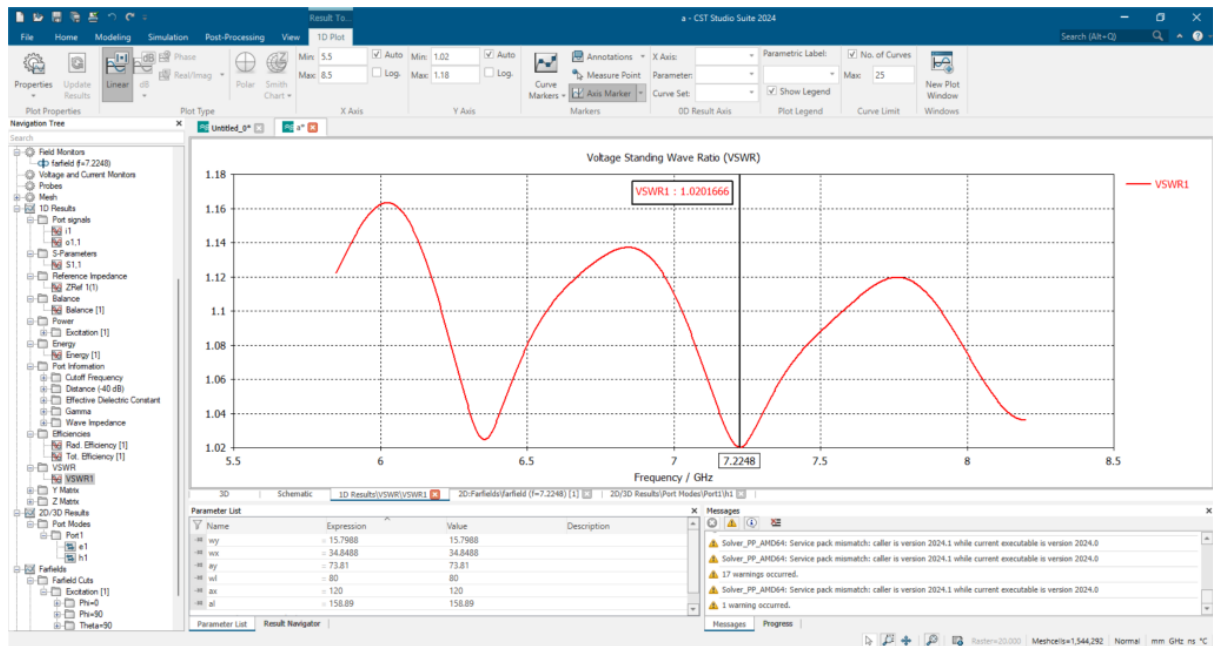


Fig 3.1 Voltage Standing Wave Ratio (VSWR) found minimum at $f = 7.2248$ GHz

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Farfield Characteristics

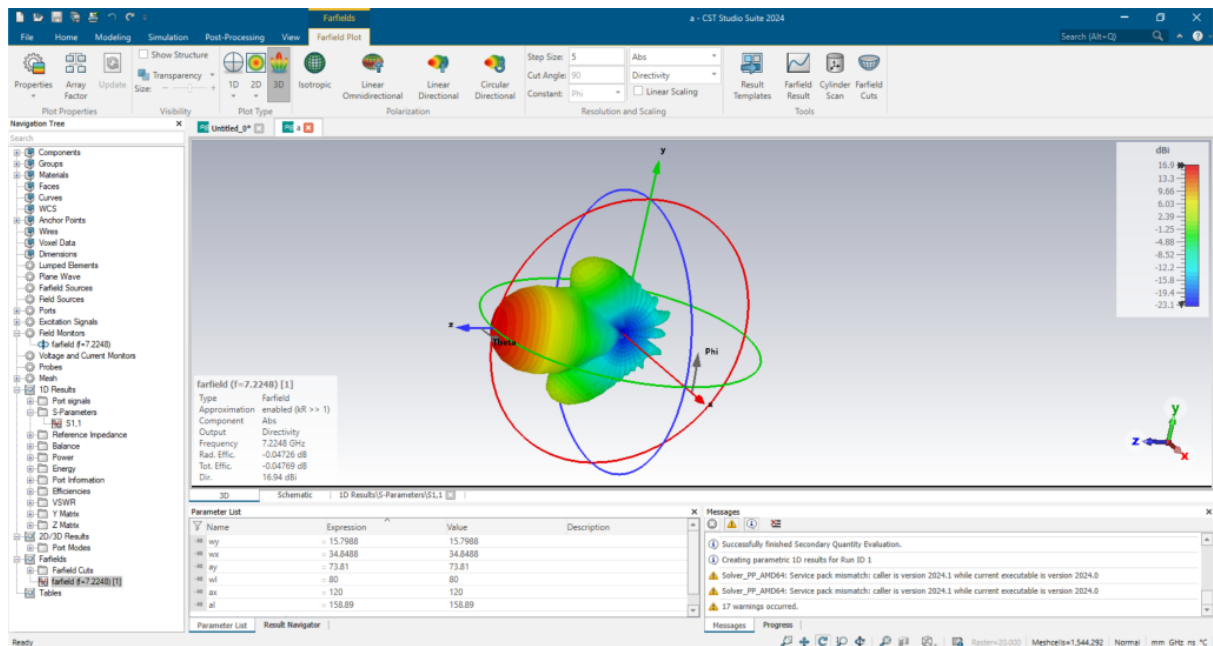


Fig 4.1 3D view of Farfield Distribution at resonant frequency $f = 7.2248$ GHz

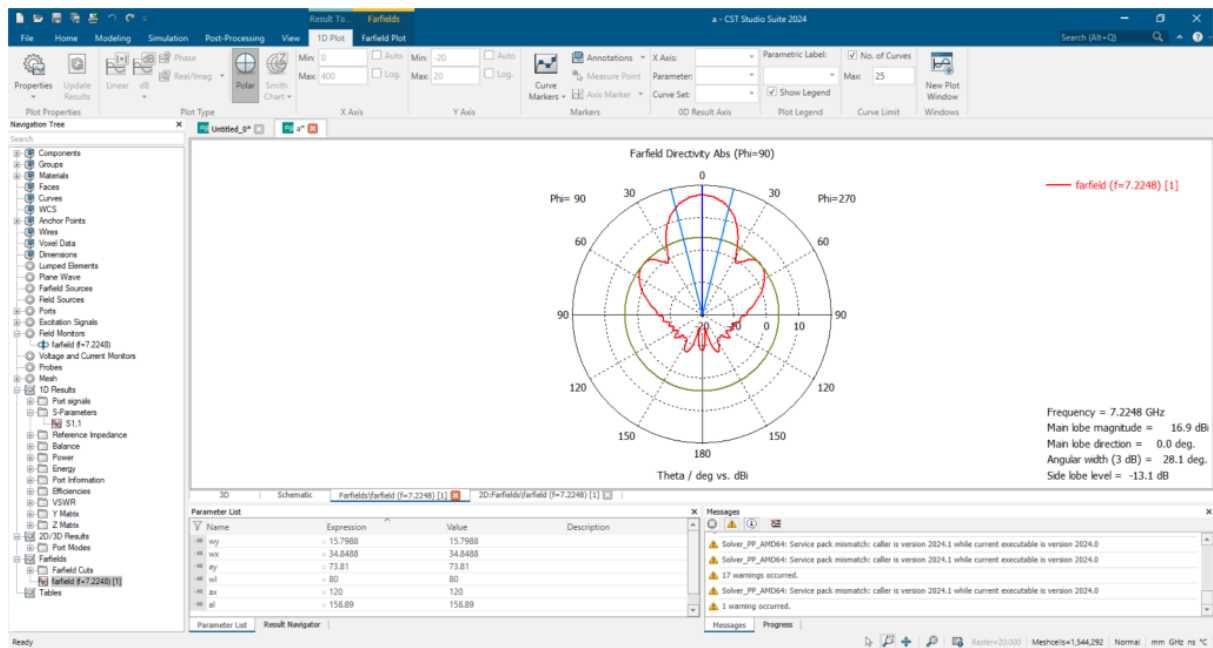


Fig 4.2 1D view of Farfield Distribution at resonant frequency $f = 7.2248$ GHz

Observations

The simulation of the pyramidal horn antenna across the standard WR137 operating band (**5.85 to 8.20 GHz**) yielded a highly favorable impedance match. Analysis of the S11 scattering parameter, which quantifies the reflection coefficient at the waveguide feed port, confirmed the antenna's broadband nature. The point of optimal resonance—defined here as **the frequency of least reflection—was observed at exactly 7.2248 GHz**. This slight upward shift from the initial 7 GHz design frequency is a standard geometric consequence of the physical flare transition, indicating the exact point where the characteristic impedance of the waveguide is most perfectly matched to the intrinsic impedance of free space.

Evaluation of the farfield radiation monitors provided critical insights into the antenna's spatial power distribution. At the optimal operating frequency, the structure produced a highly focused, directional main lobe characteristic of a dual-plane flared geometry. **The observed maximum directivity was recorded at 16.94 dBi**. When compared against the theoretical baseline target of 18 dBi, which assumes a mathematically perfect uniform phase distribution across the aperture, this result translates to a realistic and highly efficient aperture efficiency of approximately 78 percent.

This reduction from the theoretical maximum to the observed 16.94 dBi is an expected physical reality of pyramidal horn design. Because the electromagnetic waves must travel from a centralized waveguide feed to an expanded rectangular **aperture of 120 mm×73.81 mm**, a spherical phase taper naturally develops. The waves reaching the extremities of the aperture are slightly out of phase with those at the center. However, because the axial length ($a_l=158.89$ mm) successfully restricted the half-flare **angle to a maximum of 15 degrees**, this phase error was kept well within acceptable operational limits, resulting in a successful and well-optimized final design.

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Conclusions:

The design and simulation of the WR137 pyramidal horn antenna successfully demonstrated the principles of broadband impedance matching and highly directive free-space radiation. Building upon the fundamental rectangular waveguide propagation characteristics explored in earlier component analyses, such as E-plane and H-plane Tee junctions, this project effectively transitioned guided electromagnetic waves into a focused, radiating beam.

The antenna was designed for a 7 GHz center frequency with a target theoretical gain of 18 dBi. By calculating the required physical aperture (120 mm×73.81 mm) and strictly constraining the half-flare angle to a maximum of 15 degrees, an optimal axial length of 158.89 mm was established.

Simulation within the standard 5.85 to 8.20 GHz operating band validated these derived dimensions. The structure achieved an excellent broadband match, with the absolute lowest S11 reflection coefficient occurring at 7.2248 GHz. Furthermore, the farfield radiation analysis recorded an observed directivity of 16.94 dBi. This realistic drop from the 18 dBi theoretical maximum confirmed an actual aperture efficiency of approximately 78 percent, perfectly illustrating the standard physical trade-off caused by spherical phase taper across the physical aperture of a pyramidal horn.

Ultimately, the model proved to be highly efficient and well-optimized, successfully acting as an impedance transformer between the WR137 feed and free space.